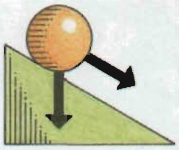
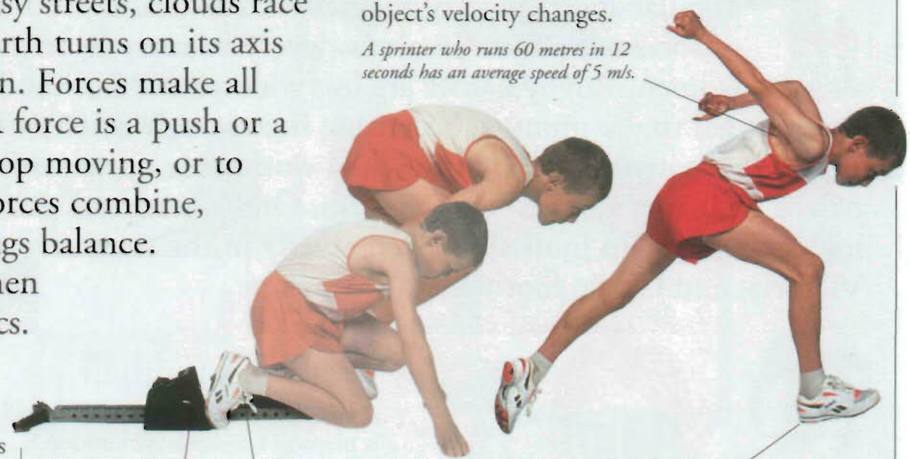
FORCE AND MOTION



THE WORLD IS NEVER STILL – traffic and pedestrians rush along busy streets, clouds race across the sky, and the Earth turns on its axis and whirls around the Sun. Forces make all this motion, or movement, possible. A force is a push or a pull that causes an object to start or stop moving, or to change its speed or direction. When forces combine, they can hold things still or make things balance. The study of the way objects move when forces act upon them is called dynamics.

Speed and acceleration

An object's speed is how far it moves in a period of time. Speed in a particular direction is called velocity. Acceleration is the rate at which an object's velocity changes.



A sprinter who runs 60 metres in 12 seconds has an average speed of 5 m/s.

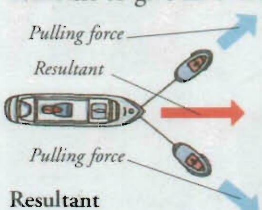
The sprinter's feet push against the starting blocks.

The force exerted by the sprinter's feet propels him forward.

A sprinter's acceleration is greatest during the first few seconds of a race.

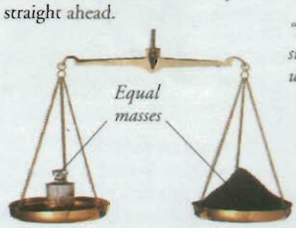
Combining forces

Equal forces acting on an object in opposite directions will have no effect. If the forces are not equal, or if they do not act in opposite directions, they will combine to give an overall force called the resultant.



Resultant
Two tugboats helping an ocean liner into port do not pull in the direction the ship needs to travel. They pull at an angle to each other so that the resultant force moves the ship straight ahead.

Terminal velocity
Gravity pulls a parachute downwards, but air resistance pushes upwards with an equal force. There is no resultant, because the forces cancel each other out. The parachute cannot accelerate, so it falls to the ground at a constant speed, known as terminal velocity.



Equilibrium
An object is in equilibrium when the forces acting upon it balance. This set of scales is in equilibrium when two equal masses are placed on the pans, because gravity pulls on each pan with the same force.

"Flying buttresses" support the walls.

Statics
Statics is the study of forces acting on stationary objects in equilibrium. It is important in building design, because a building will collapse if the forces acting upon it do not balance.

Circular motion

A free-moving object will naturally move in a straight line. Centripetal force is needed for the object to move in a circle. This is a force that pulls an object towards the centre of a circle, constantly changing its direction and stopping it from moving off in a straight line. A motorcycle uses centripetal force to travel around a bend.

Friction between the tyres and the road provides centripetal force.

Inertia

An object's mass makes it resist a force that tries to change its state of motion, whether it is moving or at rest. This resistance is called inertia. The greater an object's mass, the more inertia it has. For example, the same force will accelerate a small car more than a loaded truck, because the car has a smaller mass and less inertia.

Momentum
When a moving object collides with a stationary one, the result depends upon a quantity called momentum. An object's momentum is calculated by multiplying its mass by its velocity. For example, a heavy bowling ball has more momentum than a light plastic ball moving at the same velocity, because it has a greater mass.

The mass of the bowling ball gives it enough momentum to scatter the pins.

The lighter plastic ball has a smaller mass and simply bounces off the pins.

Archimedes

Archimedes (c.287–212 BC) was a Greek mathematician and inventor who studied forces and how they could be used by simple machines. He founded statics, discovered why objects float and sink, and worked out the principles behind levers and pulleys.

Newton's laws of motion

In 1687, English physicist Sir Isaac Newton devised three laws to summarize the principles of force and motion.

Force → Motion →

First law
An object continues in a state of rest or constant motion unless a force acts upon it. The inline skater in the picture will keep on rolling at the same speed until a force, such as friction, acts to stop him.

Second law
An object's acceleration is equal to the size of the force acting upon it divided by the object's mass. This inline skater's acceleration depends on how heavy he is and how hard he is pushed.

Third law
For every force there is an equal force acting in the opposite direction. Forces act in pairs, so when A pushes B, an equal and opposite force acts on A, making both inline skaters move apart.

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